

Chapter 8 - Inverse Circular Functions and Trigonometric Equation

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#Circular Functions

The function of an angle defined as the ratio between the sides of a right angled triangle in a unit circle is called circular/trigonometric function. There are 6 trigonometric functions. Among them, most used are:

1. **Sine function:** $y = \sin x$
2. **Cosine function:** $y = \cos x$
3. **Tangent function:** $y = \tan x$

#Inverse Circular Functions

The function obtained by interchanging the elements of domain and range of a circle function is called inverse circular function. E.g., $f(x) = \sin x \Rightarrow f^{-1}(x) = \sin^{-1} x$. [**Note:** The notation $\sin^{-1} x$ is not same as $(\sin x)^{-1}$.]

Each trigonometric function has its inverse function, but we'll only study the following:

Function	Domain	Range
$y = \sin^{-1} x$	$[-1, 1]$	$[\frac{-\pi}{2}, \frac{\pi}{2}]$
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$y = \tan^{-1} x$	R	$(\frac{-\pi}{2}, \frac{\pi}{2})$
$y = \cot^{-1} x$	R	$(0, \pi)$

Some Useful Results

Inverse Function Property:

- $\sin(\sin^{-1} x) = \sin^{-1}(\sin x) = x$
- $\cos(\cos^{-1} x) = \cos^{-1}(\cos x) = x$

This property is shown by all remaining functions.

Reciprocal Relations:

- $\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$
- $\cot^{-1} x = \tan^{-1} \frac{1}{x}$
- $\sec^{-1} x = \cos^{-1} \frac{1}{x}$

Relation with Others:

- $\sin^{-1} x = \cos^{-1} \sqrt{1-x^2}$
- $\sin^{-1} x = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$
- $\sin^{-1} x = \cot^{-1} \frac{\sqrt{1-x^2}}{x}$
- $\sin^{-1} x = \sec^{-1} \frac{1}{\sqrt{1-x^2}}$
- $\sin^{-1} x = \operatorname{cosec}^{-1} \frac{1}{x}$

Complementary Relations:

- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
- $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$

Negative Input:

- $\sin^{-1}(-x) = -\sin^{-1} x$
- $\cos^{-1}(-x) = \pi - \cos^{-1} x$
- $\tan^{-1}(-x) = -\tan^{-1} x$
- $\cot^{-1}(-x) = \pi - \cot^{-1} x$

#Introduction of Trigonometric Equations

A trigonometric equation is an equation involving one or more trigonometric functions of a variable. This equation may be true for one or more values of the variable.

The simplest type of trigonometric equation is that in which a trigonometric function of a variable (angle) is equal to a constant. Since the trigonometric function is a cyclic function so there exists an infinite number of roots of the equation.

General Solution of Trigonometric Equation

A trigonometric equation has multiple solutions. The solutions differ by an integral multiple of 2π . The angles formed by addition and subtraction of 2π are called coterminal angles. The set of all possible solutions of a trigonometric equation form the general solution of the equation. The general solutions of some trigonometric functions are tabulated below:

General Solution

1. $\sin x = k \Rightarrow x = n\pi + (-1)^n \sin^{-1} k, (n \in N)$
2. $\cos x = k \Rightarrow x = 2n\pi \pm \cos^{-1} k, (n \in N)$
3. $\tan x = k \Rightarrow x = n\pi \pm \tan^{-1} k, (n \in N)$

Reciprocal Solution

1. $\operatorname{cosec} x = k \Rightarrow x = n\pi + (-1)^n \operatorname{cosec}^{-1} k, (n \in N)$
2. $\sec x = k \Rightarrow x = 2n\pi \pm \sec^{-1} k, (n \in N)$
3. $\cot x = k \Rightarrow x = n\pi + \cot^{-1} k, (n \in N)$

Special Solution

1. When $k = 0$
 - a. $\sin x = 0 \Rightarrow x = n\pi, (n \in N)$
 - b. $\cos x = 0 \Rightarrow x = (2n + 1)\frac{\pi}{2}, (n \in N)$
 - c. $\tan x = 0 \Rightarrow x = n\pi, (n \in N)$
2. When $k = 1$
 - a. $\sin x = 1 \Rightarrow x = n\pi + \frac{\pi}{2}, (n \in N)$
 - b. $\cos x = 1 \Rightarrow x = 2n\pi, (n \in N)$
 - c. $\tan x = 1 \Rightarrow x = n\pi + \frac{\pi}{4}, (n \in N)$
3. When $k = -1$
 - a. $\sin x = -1 \Rightarrow x = n\pi - \frac{\pi}{2}, (n \in N)$
 - b. $\cos x = -1 \Rightarrow x = (2n + 1)\pi, (n \in N)$
 - c. $\tan x = -1 \Rightarrow x = n\pi - \frac{\pi}{4}, (n \in N)$

Trigonometric Equations in Other Forms.

If the given trigonometric equation is not in the simplest form, we solve it partially by using quadratic equation's general solution or by factorisation. The below hints helps in solving a trigonometric equation:

1. Express, if possible, all trigonometric functions in terms of a single trigonometric function of the same angle.
2. Transfer every term to the left.
3. If 2nd degree is obtained in a certain trigonometric function, use the formula for the solution of a quadratic equation. Or, factorise the left side, and equate each factor to zero.
4. Use the general values of the last section.

Note: In the process of solving trigonometric equations, we may obtain some additional roots. So, we should check all the obtained roots if they satisfy the initial (given) equation.